CS & IT ENGINEERING

COMPILER DESIGN

PARSING

Lecture -06

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Recap of Previous Lecture

Topic ?? LR (0) Parsing

SLR (1) Parsing follow()

Topics to be Covered

Topic ??

Topic ??

Topic ??

Parsing

R/R Conflict

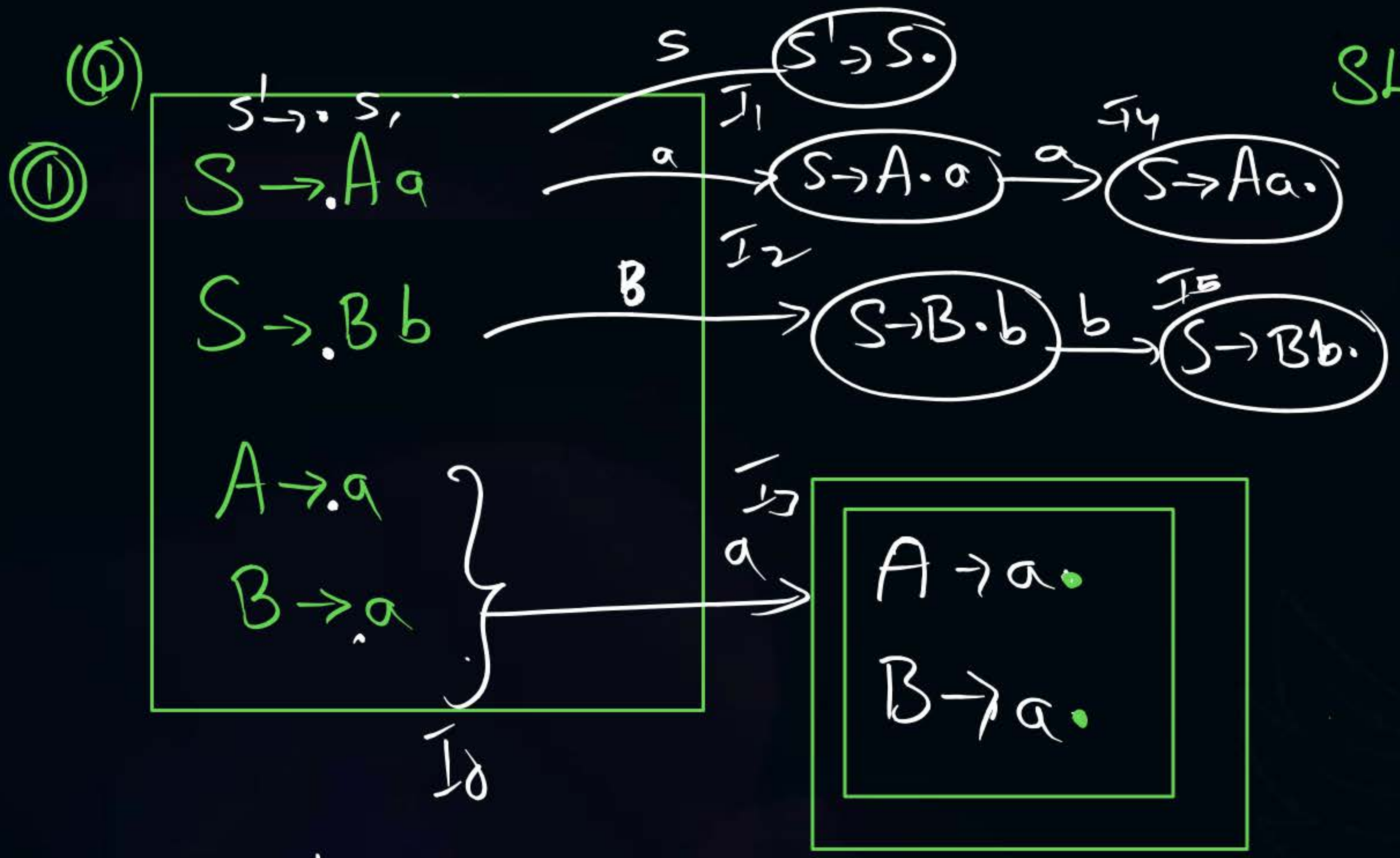
I_m

$A \rightarrow \alpha\cdot$

① $B \rightarrow \gamma\cdot$

② $Follow(A) \cap Follow(B) \neq \phi$

SLR(1) Grammar (or) not?



SLR(1) Grammar

LR(0) items DFA

$\underbrace{\text{Follow}(A)}_{\{a\}} \cap \underbrace{\text{Follow}(B)}_{\{b\}} = \emptyset$ } No R/R Conflict

SLR(1) Grammar or not ?

(2)

$$S' \to \cdot S$$
$$S \to \cdot A$$
$$S \to \cdot a$$
$$A \to \cdot a$$

$$\xrightarrow{a}$$

$$S \to a\cdot$$
$$A \to a\cdot$$

R|R Conflict

Not SLR(1) Grammar

$$\underset{\{\$\}}{\text{Follow}(S)} \cap \underset{\{\$\}}{\text{Follow}(A)} \neq \phi$$

Topic : Conflict States For SLR(1) Parser

S/R Conflict

terminal

① $A \to \alpha \cdot a B$ (Shift)

$B \to \gamma \cdot$ (Reduce)

② if $a \in \boxed{\text{Follow}(B)}$ then S/R Conflict ✓

SLR(1) Grammar(or) not?

$S' \to .S$

$S \to .AaBb$

$S \to .BbBa$

① $\begin{cases} A \to .\epsilon. \\ B \to .\epsilon. \end{cases}$

{NOT SLR(1) Grammar}

If $\underbrace{Follow(A)}_{\{a\}} \cap \underbrace{Follow(B)}_{\{a,b\}} \neq \phi$ } R/R Conflict

$S \rightarrow A$

$A \rightarrow AB$

$A \rightarrow \epsilon$

$B \rightarrow aB$

$B \rightarrow b$

SLR(1) Grammar (or) not ?

Home Work

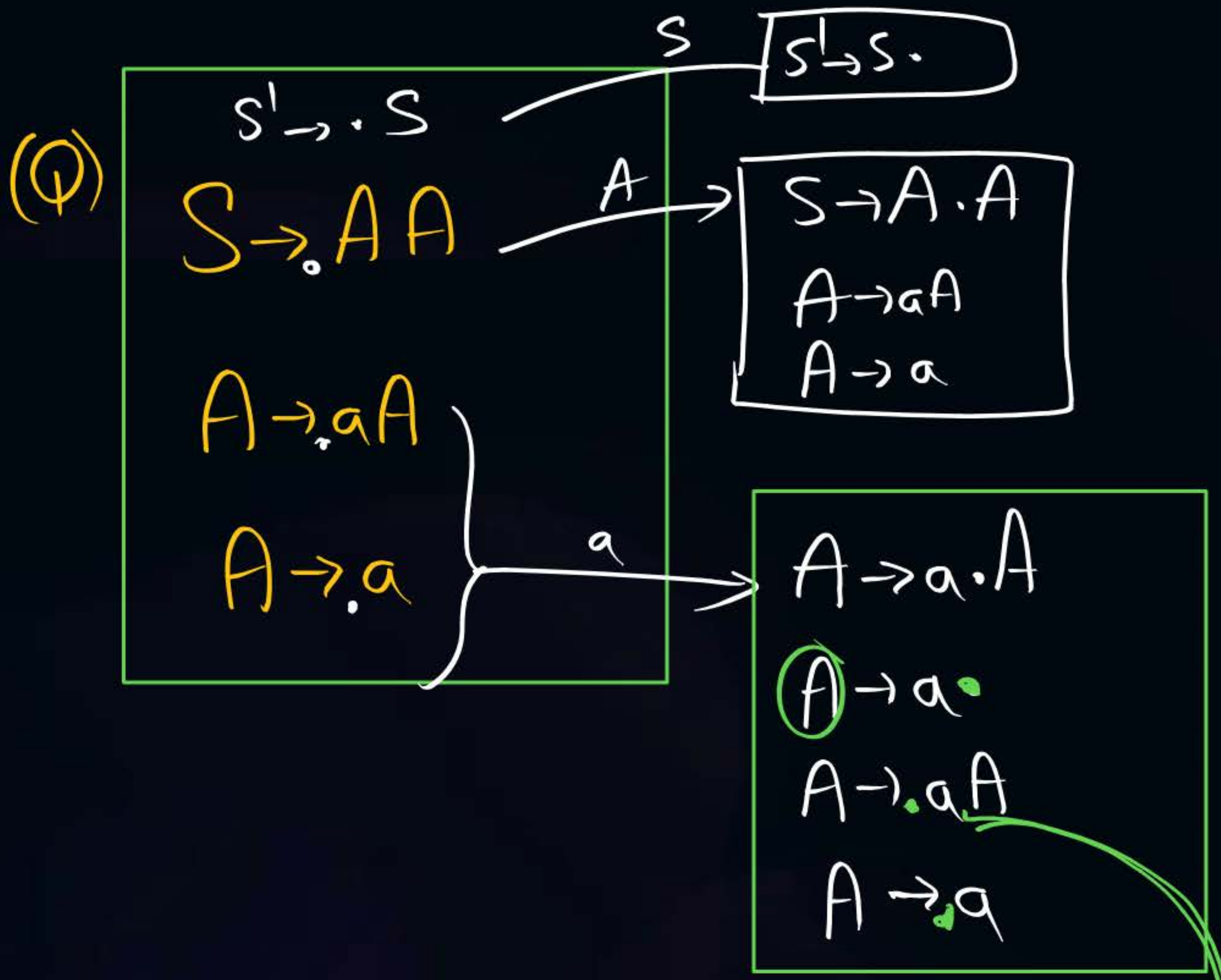
Which of the following is accepted by DPDA

~~(a) Regular~~

(b) CFL

(c) NPDA language

(d) decidable language



not SLR(1) Grammar

S/R Conflict

Follow(A) = {a, $}

② $S \to Aa \mid bAc \mid dc \mid bda$

$A \to d$

SLR(1) Grammar (or) not?

$S' \to \cdot S$

$S \to \cdot Aa$

$S \to \cdot bAc$

$S \to \cdot dc$

$S \to \cdot bda$

$A \to \cdot d$

$\xrightarrow{d}$

$S \to d \cdot c$

$A \to d \cdot$

Follow(A) = $\{a, c\}$

S/R conflict exist

not SLR(1) Grammar

#Q. Which of the following grammar is SLR(1) grammar?

Canonical LR(1) [CLR(1)]

1. Augmented Grammar

2. Construct $\boxed{\text{LR(1) items DFA}}$ → $\boxed{\text{closure()}}$ ✓, → $\boxed{\text{goto()}}$

   LR(0)

3. Construct CLR(1) Parsing table

Closure( )

$$\left\{\begin{array}{l} S' \to S \\ S \to AB \\ A \to a \\ B \to b \end{array}\right\}$$

closure( )

$S' \to \cdot \boxed{S}\ \{\$\}$

$S \to \cdot AB$, First(\$) = {\$}

$A \to \cdot a$, First(B) = {b}

closure( )

$S \to A \cdot B, \$$

$B \to \cdot b$, First(\$) = {\$}

goto()

$$\boxed{S \to A \cdot B, \$} \xrightarrow{goto()} \boxed{S \to AB \cdot, \$}$$

(Q) Construct LR(1) items DFA for given grammar

$S \rightarrow CC$

$C \rightarrow aC$

$C \rightarrow d$

10 (1) Number of States of Parsing Table

(2) Number of Shift actions

(3) " " Reduce actions

(4) Conflicts

(Q) Construct LR(1) items DFA for given grammar

① $S' \to S$

$S \to (S)$

$S \to a$

find number of states in LR(1) items DFA ?

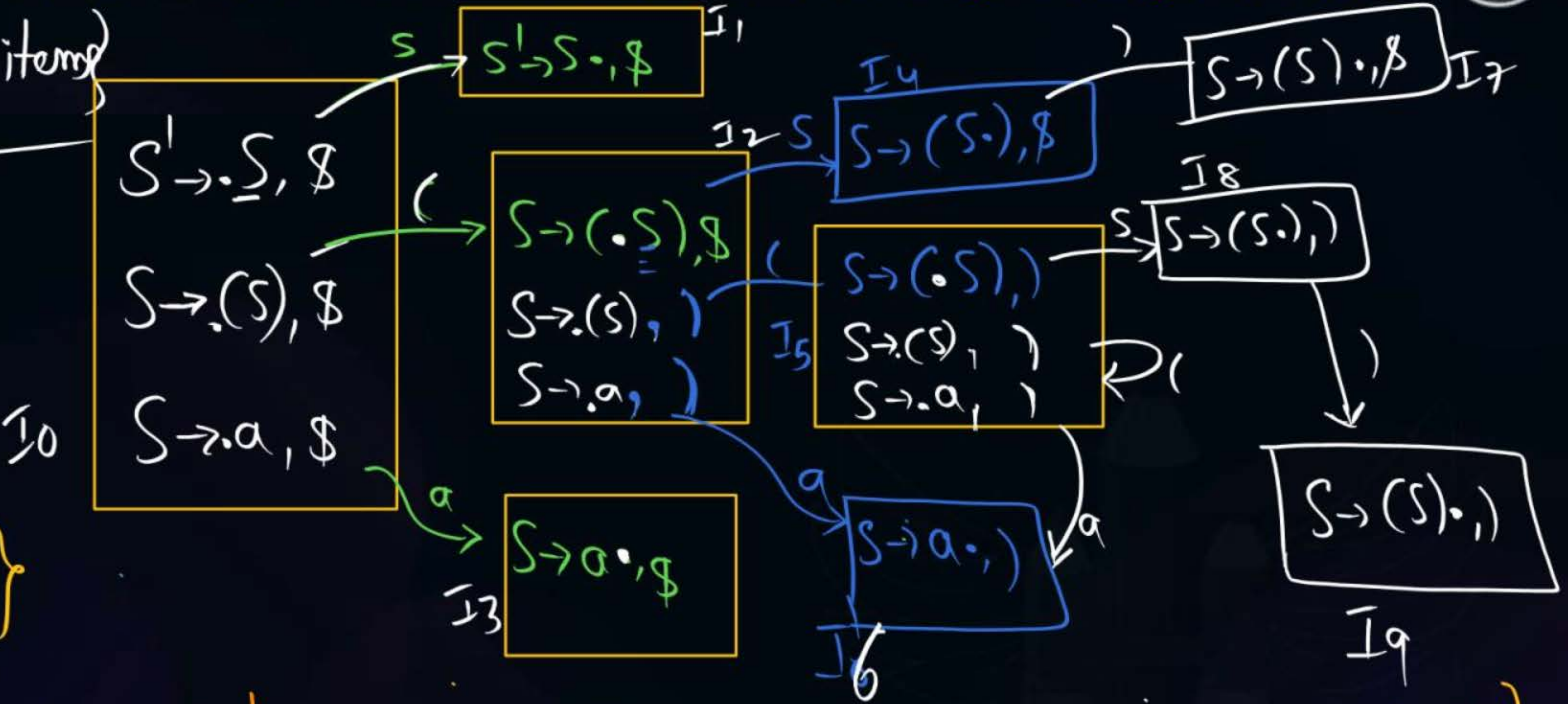
(Q) Construct LR(1) items DFA for given grammar

{LR(1) items} DFA

$S \to (S)$

$S \to a$

$\{a, (,)\}$



10 States

Size of CLR(1) Parsing Table $= 10 \times (3+1+1)$

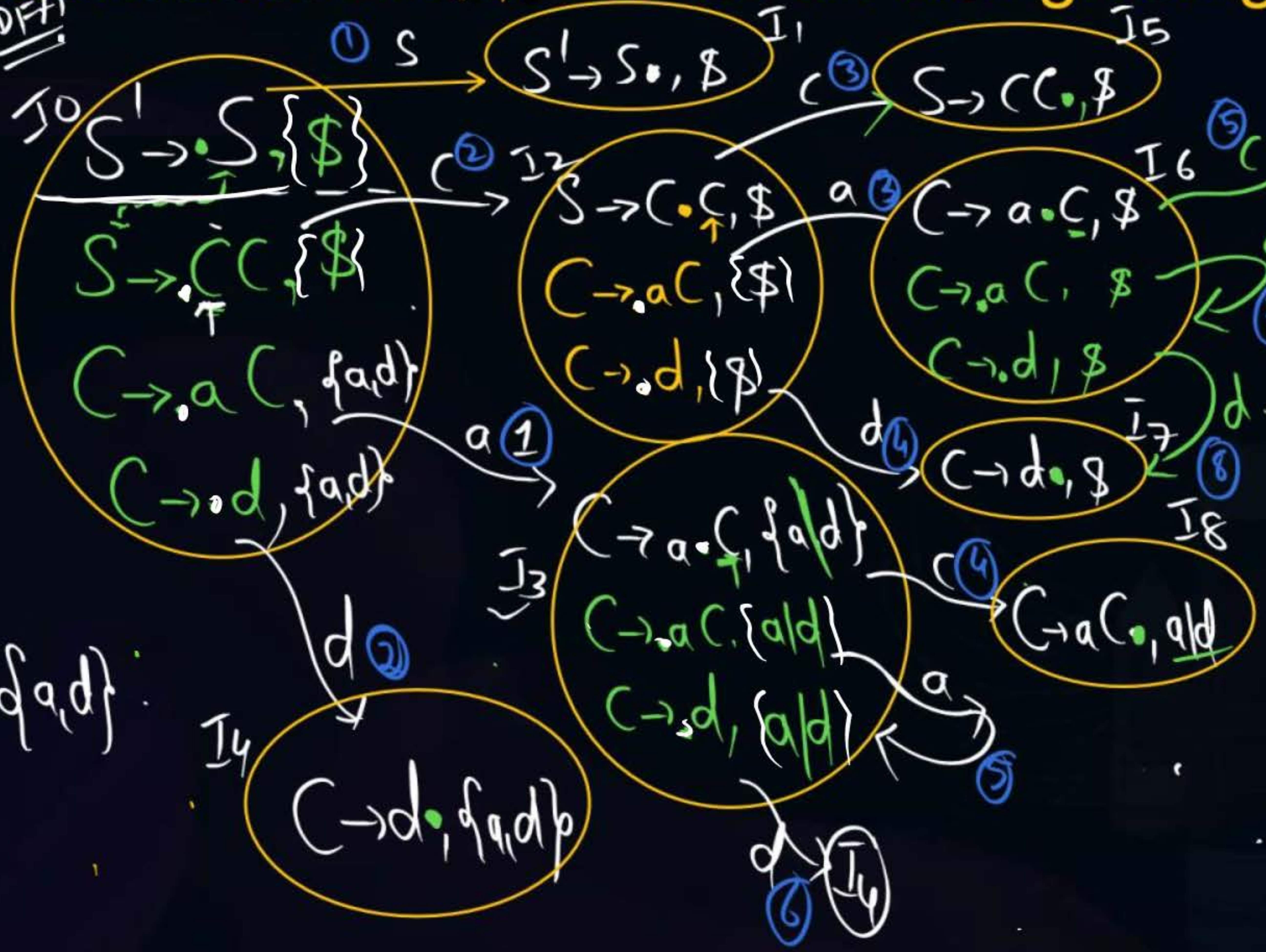
$10 \times 5 = 50$

(Q) Construct LR(1) items DFA for given grammar

LR(1) items DFA

{a, d}

$$\left\{\begin{array}{l} S \to CC \\ C \to aC \\ C \to d \end{array}\right\}$$



First(C) = {a, d}

δ Shift action

[terminal transition]

Shift actions

GOTO (5 entry)

2 mins Summary

Topic One

Topic Two

Topic Three

Topic Four

Topic Five

THANK - YOU